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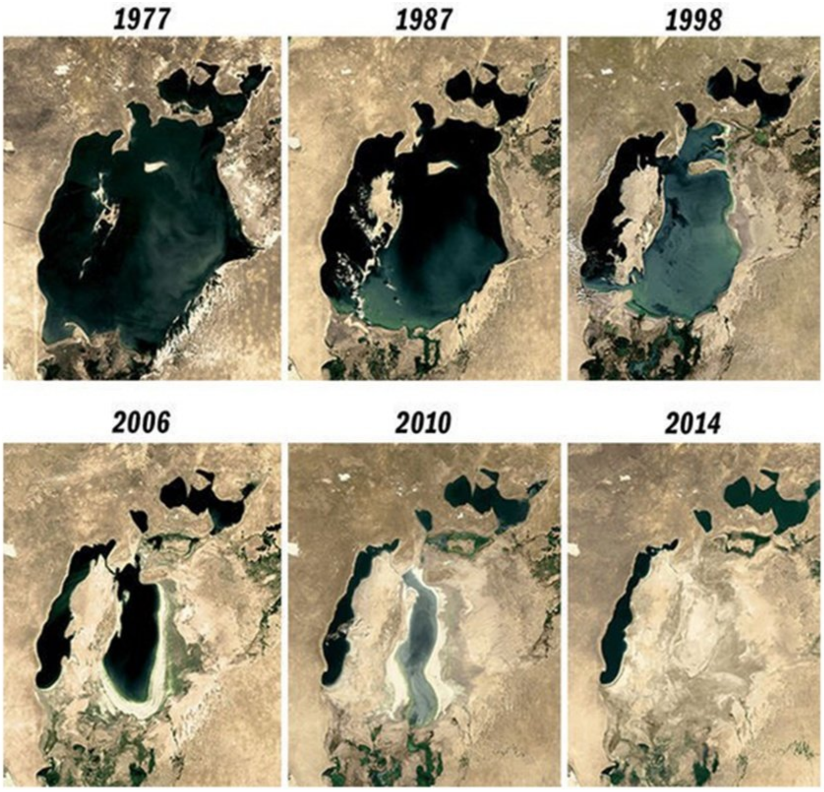
THE HUMANITARIAN SITUATION OF THE POPULATION OF KARAKALPAKSTAN IN THE CONTEXT OF THE ARAL SEA ECOLOGICAL CATASTROPHE

An Appeal to the International Community

To the United Nations (UN) and its specialized agencies;
To the Organization for Security and Co-operation in Europe (OSCE);
To the Office of the UN High Commissioner for Human Rights (OHCHR);
To diplomatic missions;
To human rights organizations and independent journalists.



The Aral Sea: 1977–2014 (satellite imagery)

June 3, 2026

Executive Summary

The desiccation of the Aral Sea stands as one of the most severe anthropogenic ecological catastrophes of the 20th century, frequently characterized by scientists as a "quiet Chernobyl." The primary burden of this disaster is borne by the population of Karakalpakstan—approximately 2 million people—inhabiting the southern coast of the former sea.

The ramifications of this crisis transcend geographical borders. Salt and toxic dust uplifted from the exposed seabed disperse over thousands of kilometers, accelerating the melting of regional mountain glaciers. Furthermore, scientific evidence indicates that the displacement of such an immense volume of water has impacted sub-surface geophysical dynamics, inducing changes within the Earth's mantle. This phenomenon constitutes an issue of global, planetary significance.

The data and evidence presented in this appeal do not constitute abstract threats; rather, they represent quantified phenomena substantiated by international scientific sources. These include the widespread prevalence of anemia, an ongoing tuberculosis epidemic, elevated rates of infant and maternal mortality, the contamination of drinking water with toxic substances, and the destruction of the fundamental livelihoods of millions of people.

We call upon the international community to direct its attention to this ongoing humanitarian crisis. We urgently request the implementation of independent ecological and medical monitoring, the assurance of transparency and accountability in the allocation of international funds dedicated to the Aral Sea region, and the provision of targeted assistance to the affected population.

I. Scale of the Disaster

The Aral Sea was once the world's fourth-largest inland body of water, covering an area of approximately 68,000 square kilometers. Beginning in the 1960s, following the diversion of the Amu Darya and Syr Darya rivers for cotton irrigation, the sea began to desiccate rapidly. By 2007, it had lost approximately 90 percent of its volume, leaving only about 10 percent of its original surface area.

The desiccation of the sea has given rise to one of the world's youngest deserts—the Aralkum Desert, which spans approximately 60,000 square kilometers. Since 1960, the sea has lost more than 1,000 cubic kilometers of water.

II. Toxic Dust: A Threat That Knows No Borders

The exposed seabed of the Aral Sea has become a repository of salts, mineral deposits, pesticides, and herbicides that accumulated over decades through agricultural runoff from cotton fields. Today, the dried seabed serves as a source of dispersal for these hazardous substances across vast areas of the region.

According to scientific estimates, between 15 and 75 million tons of salt and toxic particulate matter are carried annually by wind from the exposed seabed. Some studies estimate that this figure exceeds 100 million tons per year. Air quality degradation has been documented at distances of up to 800 kilometers from the source.

This is not ordinary desert dust. Scientific research has detected elevated concentrations of DDT, arsenic, mercury, and other toxic organochlorine compounds in the blood, urine, and even breast milk of residents of the Aral Sea region.

The impact of this dust extends far beyond the borders of Karakalpakstan. Scientific studies and atmospheric modeling indicate that dust originating from the Aral Sea basin can travel hundreds and even thousands of kilometers, reaching Georgia and the Arctic, while some models suggest the potential for transport as far as Greenland. Of even greater concern, the deposition of toxic dust on glacier surfaces and the increased mineralization of precipitation accelerate the melting of mountain glaciers that feed the Amu Darya and Syr Darya river systems, thereby exacerbating future water scarcity throughout the region.

III. Impact on Public Health

The humanitarian dimension of the disaster is most clearly reflected in its effects on public health.

Anemia. According to studies based on data from the World Health Organization (WHO), the prevalence of anemia among women and children in Karakalpakstan ranges between 80 and 90 percent, placing it among the highest rates recorded anywhere in the world. Some studies indicate that the prevalence among pregnant women reaches as high as 99 percent.

Tuberculosis. The World Health Organization identifies 50–70 cases per 100,000 people as an epidemic threshold. In Karakalpakstan, the incidence rate is approximately 220 cases per 100,000 population. The region is also recognized as having one of the highest rates of multidrug-resistant tuberculosis (MDR-TB) in the world.

Infant and Maternal Mortality. Scientific studies report infant mortality rates in Karakalpakstan ranging from 60 to 110 deaths per 1,000 live births, substantially exceeding both the average rate for Uzbekistan (approximately 48 per 1,000) and that of Russia (approximately 24 per 1,000).

Respiratory Diseases, Cancer, and Congenital Disorders. Exposure to toxic dust has been associated with a significant increase in chronic bronchitis, asthma, kidney and liver diseases, cancer, and congenital abnormalities. Acute respiratory illnesses account for nearly half of all child mortality cases in the region.

IV. Socio-Economic Consequences

The fishing economy that once sustained tens of thousands of people has completely collapsed; port cities such as Moynaq now lie hundreds of kilometers away from the former shoreline. According to expert assessments, the disaster has forced more than 100,000 people to leave their homes and has affected the health of over 5 million people across the wider region. Remaining water sources have been contaminated by agricultural chemicals, and the quality of drinking water has reached a critical level.

V. Planetary and Historical Significance of the Catastrophe

Impact on the Earth's mantle. A study published in 2025 in Nature Geoscience (Peking University, University of Southern California) found that, as a result of the Aral Sea's disappearance, the exposed seabed is rising by approximately 7 millimeters per year within a radius of 500 kilometers. The cause is the "rebound" of the Earth's crust after being relieved of the weight of more than 1,000 cubic kilometers of water. The authors conclude that human activity can influence even the deep internal dynamics of the Earth. This phenomenon clearly demonstrates the planetary scale of the disaster.

Legacy of the biochemical testing site. Between 1942 and 1992, the Soviet biochemical testing ground ("Barkhan," PNIL-52) operated on Vozrozhdeniye (Barsakelmes) Island in the Aral Sea. As the sea receded, the site became part of the mainland, creating an additional threat to the region's environmental security.

VI. Uzbekistan's Water Policy: Transboundary Damage and Potential Intentional Objectives

Deliberate Blockage of the Amu Darya River. International scientific monitoring and local-level data indicate that Uzbekistan is deliberately blocking the waters of the Amu Darya, preventing them from reaching the Aral Sea. This action is evaluated as serving two potential objectives:

1. Drying up the sea bed to facilitate the extraction of natural resources (gas, minerals, and other deposits) through simplified access. The resource potential beneath the Aralkum Desert is scientifically estimated to be substantial.
2. Inducing a severe water shortage for the population of Karakalpakstan to compel displacement, thereby establishing direct ownership over the land and its resources. The 2022 attempt to revoke the sovereign status of Karakalpakstan is cited as evidence of this policy.

Transparency of Reservoirs. Official documentation (UNFCCC) lists 70 registered artificial reservoirs within Uzbekistan. However, local assessments place this figure at over 140–150. A significant number of these reservoirs are not visible on standard public mapping services (such as Google Maps), suggesting clandestine operation. Consequently, it is imperative that the United Nations establishes an independent, credible commission to conduct an on-site, rigorous verification process within Uzbekistan.

VII. International Legal Framework

This matter is recognized at the international level and transcends domestic jurisdiction based on the following frameworks:

1. On May 18, 2021, the UN General Assembly unanimously adopted Resolution A/RES/75/278, declaring the Aral Sea region a zone of ecological innovations and technologies.
2. In 1993, regional states established the International Fund for Saving the Aral Sea (IFAS), and the UN system subsequently operated the Multi-Partner Human Security Trust Fund for the Aral Sea Region.

3. On July 28, 2022, the UN General Assembly adopted Resolution A/RES/76/300, recognizing the right to a clean, healthy, and sustainable environment as a universal human right. The situation in the Aral Sea region directly intersects with this right.

This issue cannot be dismissed as an internal matter of Uzbekistan. First, the transboundary dispersal of dust and toxic substances impacts multiple states, constituting transboundary environmental damage. Second, human rights—specifically the right to health and a healthy environment—are subject to international scrutiny and cannot be classified exclusively as domestic affairs. On these two grounds, the Aral crisis remains a legitimate subject of international oversight and concern.

VIII. Historical and Legal Reminder: Restoring the Sovereign Status of Karakalpakstan

On 14 December 1990, the Supreme Council of the Karakalpak ASSR (Resolution No. 82/XII, Chair T. Yeshimbetova) adopted the Declaration on State Sovereignty. Its preamble states:

> *"Committed to the development of peoples and to addressing the environmental problems arising from the drying of the Aral Sea, taking into account the extremely low living standards of citizens residing in the zone of ecological disaster…"*

The Declaration is a historic document that defined the political and legal status of Karakalpakstan. Moreover, Karakalpakstan is recognized as a sovereign republic within the Constitution of Uzbekistan itself.

Proposal to the international community: To restore the sovereign status of Karakalpakstan, as established on 14 December 1990, in accordance with international law. This proposal aligns with current international practice, as demonstrated by the example of Latvia. It may serve as an effective and just pathway toward resolving one of the world's most severe crises. The voice of the region that has suffered the greatest harm must be heard directly.

IX. Demands and Recommendations

Based on the facts outlined above, we request the following from the international community:

1. Independent monitoring. Establish independent environmental and public-health monitoring in Karakalpakstan with the participation of international experts; if necessary, deploy a UN technical mission.
2. Accurate assessment of water resources. Under UN auspices, form a credible commission to conduct an exact inspection of the true number of reservoirs in Uzbekistan. It must be verified whether the actual number exceeds 140–150, rather than the officially stated 70.
3. Transparency and accountability of financial flows. Publish open, audited reports on how international funds and grants related to the Aral Sea are allocated and spent; ensure that assistance reaches the affected population.

4. Targeted humanitarian programmes. Support targeted initiatives providing clean drinking water, treatment for tuberculosis and anemia, and protection of maternal and child health.
5. Local voice and press freedom. Ensure the participation of local representatives and journalists in decision-making processes; guarantee safe and unhindered access for observers and the press.
6. Sustained attention. Maintain this crisis under continuous international oversight rather than temporary interest.
7. Restoration of Karakalpakstan's sovereign status. Restore the sovereign status of Karakalpakstan, based on the Declaration of 14 December 1990 and in accordance with international law (as in the Latvian example). This would provide a sound legal foundation for the rapid and effective resolution of the Aral crisis.

Conclusion

The Aral catastrophe is not merely a story of a vanished sea. It is the story of millions of people forced to live each day amid disease, contaminated water and air, and the loss of their traditional way of life. Their rights to health, clean water, and a safe environment are universal human rights.

We present these facts based on evidence and reliable data, and we call on the international community for cooperation and support.

The people of Karakalpakstan deserve attention, justice, and a future.

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